



Report on Co-Curricular activities done as a part of

ACXC109N Skills in Competitive Coding

Submitted by

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To

Prof. MURALI S

Fall Semester,2025-26

Table of Activities

S. No	<Topic – It can be events/ activities/ competition anything pertaining to the course>	Date	Hours Spent	Page No.
CO 1	To apply science, engineering, management and technology to understand real world problem		Hours: 30	Marks: 30
Activities : Participation in events like orientation session, training, guest lecture, workshop & internal competitions, organized by associated technical club/ chapter/ teams				
1	Orientation	3rd September	2	4
2	Time and space complexity	6th January	2	4
3	Strings and Arrays	7th January	2	4
4	Stacks, Queues and Linked Lists	8th January	2	4
5	Trees and Heaps	9th January	2	5
6	Graphs	10th January	2	5
7	HashMaps and HashSets	11th January	2	5
8	Dynamic Programming and Greedy Algorithms	12th January	3	6
9	Bit Manipulation, Math and Geometry	13th January	3	6
10	Prefix and Suffix Arrays	14th January	2	6
11	Prime Numbers, GDC and HCF	19th January	2	7
12	SDG: TrueSight	15th October	2	7
13	CASA: Empower, Educate, Eradicate	5th November	2	8
14	Alumni Connect	14th December	2	8
CO 2	To gain practical knowledge about best practices in chosen domain		Hours: 30	Marks: 30
Activities : Gaining of practical knowledge through participation in outreach activities, field visits and relevant events				
1	Yantra 2026	8th-11th February	48	8
CO 3	To master team building and leadership skills		Hours: 30	Marks: 30

Activities: Participating / Organizing / Volunteering for events during graVITas organized by the associated chapter/ technical club thereby enhancing and exhibiting the team building and leadership skills

Representing the institute as a part of technical Club/ Chapter/ team in events or competitions conducted by other institutes

1	Vision to Visuals	22 nd January	2	9
2	React Rumble	23 rd January	2	9
3	Express Yourself	24 th January	2	9
4	Dare to Dart	3 rd February	2	10
5	Git Gud	4 th February	2	10
6	MLtiverse Of Madness	4 th February	2	10
7	Pseudo to Sudo	5 th February	2	12
8	Ready Player 1	5 th February	2	12
9	DEVSOC 25	8 th – 11 th February	48	12
10	CookOff 10.0	28 th September	8	13
11	Clueminati 3.0	25 th September	4	13
Quality of the Report (format, spacing, appropriate images), uploading on time in VTOP				Marks:10

Event Name: Orientation

➤ Description:

The orientation session introduced the course structure, objectives, evaluation pattern, and the various co-curricular activities planned for the semester. It provided a clear overview of what is expected from students and how the activities contribute to overall learning. The session was informative and helped me understand how to approach the course effectively. I learned the importance of participating in co-curricular activities and how they help in developing practical knowledge and skills.

Event Name: Time and Space Complexity

➤ Description:

The session focused on understanding time and space complexity of algorithms, including concepts like Big-O notation and different cases such as best, average, and worst-case complexity. Simple examples were used to explain how algorithm efficiency is measured. The session was easy to follow and helped clarify important concepts. I learned how to analyze algorithms and understood the importance of writing efficient and optimized code.

Event Name: Strings and Arrays

➤ Description:

This session covered the basics of strings and arrays, including operations like traversal, searching, and manipulation. Various problem-solving techniques were discussed to handle common coding problems. The session was interactive and helped me relate concepts to practical scenarios. I learned how to work efficiently with strings and arrays and improved my logical thinking and coding skills.

Event Name: Stacks, Queues and Linked Lists

➤ Description:

The session focused on fundamental data structures such as stacks, queues, and linked lists, along with their operations like insertion, deletion, and traversal. Concepts like LIFO and

FIFO were explained with examples. The session was clear and helped me understand how these structures work internally. I learned how to use these data structures effectively and understood their importance in solving programming problems.

Event Name: Trees and Heaps

➤ Description:

The session focused on advanced data structures such as trees and heaps, including concepts like binary trees, binary search trees, and heap structures. It explained how data is organized hierarchically and how heaps are used for efficient priority-based operations. The session was informative and helped me understand these concepts through simple examples. I learned how trees and heaps are applied in problem-solving and improved my understanding of hierarchical data structures.

Event Name: Graphs

➤ Description:

The session focused on graph data structures and their applications in solving real-world problems. It covered concepts such as nodes, edges, types of graphs, and traversal techniques like BFS and DFS. The session was engaging and helped me understand how graphs are used in areas like networking and pathfinding. I learned how to represent and traverse graphs effectively and

Event Name: HashMaps and HashSets

➤ Description:

The session focused on hash-based data structures such as HashMaps and HashSets, including their working principles and common use cases. It explained concepts like hashing, key-value pairs, and handling collisions with simple examples. The session was clear and helped me understand how these structures provide efficient data storage and retrieval. I learned how to use HashMaps and HashSets effectively in solving problems and improved my understanding of optimized data handling.

Event Name: Dynamic Programming and Greedy Algorithms

➤ Description:

The session focused on important problem-solving techniques such as dynamic programming and greedy algorithms. It explained how dynamic programming breaks problems into smaller subproblems and stores results to avoid recomputation, while greedy algorithms make optimal choices at each step. The session was insightful and helped me understand when to use each approach through examples. I learned how to identify patterns in problems and apply these techniques effectively, improving my problem-solving and analytical skills.

Event Name: Bit Manipulation, Math and Geometry

➤ Description:

The session focused on problem-solving techniques involving bit manipulation, mathematical concepts, and basic geometry. It covered operations like bitwise AND, OR, XOR, and shifting, along with their applications in optimizing solutions. Mathematical topics such as number properties and geometry basics were also discussed. The session was interesting and helped me understand how these concepts can simplify complex problems. I learned how to apply bitwise operations and mathematical logic to improve efficiency in coding.

Event Name: Prefix and Suffix Arrays

➤ Description:

The session focused on prefix and suffix arrays and their applications in solving range-based problems efficiently. It explained how cumulative sums can be used to reduce time complexity and improve performance in queries. The session was clear and helped me understand how to precompute values for faster calculations. I learned how to apply prefix and suffix techniques in problem-solving and improved my ability to optimize solutions.

Event Name: Prime Numbers, GCD and HCF

➤ Description:

The session focused on fundamental mathematical concepts such as prime numbers, greatest common divisor (GCD), and highest common factor (HCF). It explained methods to identify prime numbers and efficient algorithms like the Euclidean algorithm to compute GCD and HCF. The session was informative and helped me understand the importance of these concepts in problem-solving. I learned how to apply these techniques in coding and improved my mathematical and logical skills.

Event Name: SDG: TrueSight

➤ Description:

The session focused on the Sustainable Development Goals (SDG) initiative named TrueSight, highlighting the importance of using technology to address real-world societal issues. It discussed how innovation and technical solutions can contribute to solving problems related to sustainability and social impact. The session was insightful and encouraged me to think beyond academics. I learned the importance of applying technical knowledge for the betterment of society and gained awareness about real-world challenges.

Event Name: CASA: Empower, Educate, Eradicate

➤ **Description:**

The session focused on the CASA initiative aimed at creating social awareness through empowerment, education, and eradication of societal issues. It highlighted the role of students in contributing to community development and discussed various activities conducted to support social causes. The session was meaningful and encouraged active participation in outreach programs. I learned the importance of social responsibility and how technical and personal skills can be used to make a positive impact on society.

Event Name: Alumni Connect

➤ **Description:**

The Alumni Connect session focused on interaction between students and alumni to share knowledge, experiences, and career guidance. Alumni spoke about their professional journeys, challenges, and opportunities in various industries, giving insights into real-world applications of academic knowledge. The session was engaging and helped me understand different career paths and industry expectations. I learned the importance of networking, continuous learning, and how to prepare myself for future career opportunities through the experiences shared by alumni.

Event Name: Yantra 2026

➤ **Description:**

Yantra 2026 is a week-long technical fest conducted at VIT that includes various events such as workshops, hackathons, and competitions organized by different technical clubs. It provides a platform for students to explore innovation and work on real-world problem statements across domains like programming, AI, and web development. The event was highly engaging and gave me practical exposure to technical problem-solving in a competitive environment. I learned how to apply theoretical concepts in real scenarios, improved my coding and teamwork skills, and gained confidence by participating in hands-on technical activities.

Event Name: Vision to Visuals

➤ Description:

The event focused on transforming ideas into visual representations through creative thinking and design. Participants were encouraged to present their ideas using visual tools and techniques, enhancing both creativity and communication skills. The event was interactive and allowed me to explore different ways of presenting concepts visually. I learned how to effectively convey ideas through design and improved my creative and presentation skills.

Event Name: React Rumble

➤ Description:

React Rumble was a technical event focused on building user interfaces using the React framework. Participants were challenged to develop interactive and responsive web applications within a limited time. The event was engaging and allowed me to apply my frontend development knowledge in a practical setting. I learned how to structure components effectively, manage time during development, and improved my skills in React and UI design.

Event Name: Express Yourself

➤ Description:

The event focused on encouraging participants to express their ideas and thoughts through creative presentations and communication activities. It provided a platform to showcase creativity, confidence, and clarity in conveying concepts. The event was engaging and helped me step out of my comfort zone while presenting my ideas. I learned the importance of effective communication, improved my presentation skills, and gained confidence in expressing my thoughts clearly.

Event Name: Dare to Dart

➤ Description:

Dare to Dart was a technical event focused on developing applications using the Dart programming language. It involved solving challenges and building small features within a limited time, testing both coding and logical skills. The event was engaging and allowed me to explore a new programming language in a practical way. I learned the basics of Dart, improved my problem-solving skills, and gained experience in developing efficient solutions under time constraints.

Event Name: Git Gud

➤ Description:

Git Gud was a technical event focused on version control using Git and GitHub. It involved tasks such as repository management, branching, merging, and collaboration workflows. The event was interactive and helped me understand how version control is used in real-world projects. I learned how to manage code efficiently, collaborate with others using Git, and improved my understanding of development workflows.

Event Name: MLtiverse Of Madness

➤ Description:

MLtiverse Of Madness was a technical event focused on concepts related to machine learning and its applications. It involved solving problems and understanding basic models and techniques used in data analysis and prediction. The event was interesting and gave me exposure to the field of machine learning in a practical way. I learned how machine learning can be applied to real-world problems and improved my analytical and problem-solving skills.

Event Name: Pseudo to Sudo

➤ Description:

Pseudo to Sudo was a technical event focused on converting logical problem statements into executable code. It involved writing pseudocode and then implementing it into actual programs, testing both understanding and coding skills. The event was engaging and helped me bridge the gap between theory and practical coding. I learned how to structure my logic clearly before coding and improved my problem-solving and programming skills.

Event Name: Ready Player 1

➤ Description:

Ready Player 1 was an interactive technical event that involved solving challenges and completing tasks in a gamified format. It combined elements of problem-solving, logical thinking, and quick decision-making under time constraints. The event was exciting and engaging, as it encouraged active participation and teamwork. I learned how to think quickly, adapt to different challenges, and improved my problem-solving and collaboration skills.

Event Name: DEVSOC 25

➤ Description:

DEVSOC 25 is a flagship hackathon organized by the CodeChef-VIT that brings together students to collaborate, innovate, and solve real-world problems. It is typically a 48-hour event where participants form teams, brainstorm ideas, and build solutions across domains such as AI, web development, and software engineering. The event includes mentorship from industry experts, guidance sessions, and opportunities to present projects. It was an exciting and challenging experience that pushed me to think creatively and work efficiently under time constraints. I learned how to collaborate effectively in a team, apply technical knowledge to real-world problems, and gained valuable exposure to hackathon environments.



Event Name: CookOff 10.0

➤ Description:

CookOff 10.0 was a competitive programming event where participants solved coding problems within a limited time. It tested logical thinking, problem-solving ability, and speed in implementing efficient solutions. The event was challenging and engaging, as it required quick understanding of problems and accurate coding. I learned how to manage time effectively during contests, improved my coding skills, and gained confidence in solving problems under pressure.

Event Name: Clueminati 3.0

➤ Description:

Clueminati 3.0 was an interactive event that involved solving clues and challenges in a sequential manner, combining logical thinking and problem-solving skills. Participants had to analyze hints carefully and work through multiple levels to reach the final solution. The event was fun and engaging, encouraging teamwork and critical thinking. I learned how to approach problems from different perspectives, improved my analytical skills, and gained experience in collaborative problem-solving.